QUBO Formulation: Multi-Vehicle Delivery Assignment Problem

Multi-Vehicle Delivery Assignment Problem

Problem Statement. Given a set of customer jobs $\mathcal{J}$, a set of vehicles $\mathcal{V}$, and a set of candidate routes $\mathcal{R}$, each route $r \in \mathcal{R}$ is associated with a delivery cost $c_r \geq 0$, a subset of jobs $J(r) \subseteq \mathcal{J}$ that it serves, and a designated vehicle $v(r) \in \mathcal{V}$ that executes it. The goal is to select a subset of routes such that every job in $\mathcal{J}$ is covered exactly once, each vehicle is assigned at most one route, and the total delivery cost is minimized.

Decision Variables. For each candidate route $r \in \mathcal{R}$, introduce a binary decision variable $x_r \in \{0, 1\}$. The variable x_r equals 1 if route r is selected, and 0 otherwise.

QUBO Derivation. Step 1 — Objective. The original combinatorial optimization problem is expressed as:

$$\min_{\mathbf{x}} \sum_{r \in \mathcal{R}} c_r x_r. \tag{1}$$

Step 2 — Constraints. The problem is subject to two sets of constraints:

$$\sum_{r \in \mathcal{R}: j \in J(r)} x_r = 1, \quad \forall j \in \mathcal{J}, \tag{C1}$$

$$\sum_{r \in \mathcal{R}: v(r) = v} x_r \leq 1, \quad \forall v \in \mathcal{V}. \tag{C2}$$

Step 3 — Penalty Conversion. We enforce the constraints by adding quadratic penalty terms to the objective with weight $\lambda > 0$.

For the equality constraint (C1), the penalty for job j is:

$$P_j = \lambda \left(\sum_{r \in \mathcal{R}: j \in J(r)} x_r - 1 \right)^2. \tag{2}$$

Expanding the square and applying the idempotent property $x_r^2 = x_r$ for binary variables yields:

$$P_j = \lambda \left(\sum_{r \in \mathcal{R}: j \in J(r)} x_r^2 - 2 \sum_{r \in \mathcal{R}: j \in J(r)} x_r + 1 \right) \tag{3}$$

$$= \lambda \sum_{r \in \mathcal{R}: j \in J(r)} x_r - 2\lambda \sum_{r \in \mathcal{R}: j \in J(r)} x_r + \lambda \tag{4}$$

$$= -\lambda \sum_{r \in \mathcal{R}: j \in J(r)} x_r + 2\lambda \sum_{\substack{r,s \in \mathcal{R} \\ r<s \\ j \in J(r) \cap J(s)}} x_r x_s + \lambda. \tag{5}$$

For the inequality constraint (C2), we penalize the simultaneous selection of any two routes assigned to the same vehicle. The penalty for vehicle v is:

$$Q_v = \lambda \sum_{\substack{r,s \in \mathcal{R} \\ r<s \\ v(r)=v(s)=v}} x_r x_s. \tag{6}$$

This term vanishes when at most one route is selected for vehicle v, and becomes strictly positive when two or more routes are selected.

Step 4 — Combined Hamiltonian. Summing the objective and all penalty contributions gives the QUBO Hamiltonian:

$$H(\mathbf{x}) = \sum_{r \in \mathcal{R}} c_r x_r + \sum_{j \in \mathcal{J}} P_j + \sum_{v \in \mathcal{V}} Q_v. \tag{7}$$

Step 5 — Q Matrix Entries and Offset. Collecting linear and quadratic terms from $H(\mathbf{x})$ yields the closed-form expressions for the QUBO coefficients:

$$Q_{r,r} = c_r - \lambda \cdot |\{j \in \mathcal{J} : j \in J(r)\}|, \quad \forall r \in \mathcal{R}, \tag{8}$$

$$Q_{r,s} = 2\lambda \cdot |\{j \in \mathcal{J} : j \in J(r) \cap J(s)\}| + \lambda \cdot \mathbb{I}(v(r) = v(s)), \quad \forall r, s \in \mathcal{R}, r < s, \tag{9}$$

$$c = \lambda |\mathcal{J}|, \tag{10}$$

where $\mathbb{I}(\cdot)$ is the indicator function. The Hamiltonian is then written as $H(\mathbf{x}) = \mathbf{x}^\top Q \mathbf{x} + c$.

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to ensure that any violation of the constraints increases the energy more than the maximum possible reduction in the objective function. Since each route cost satisfies $c_r \geq 0$, the maximum objective gain from setting an additional variable to 1 is bounded by $\max_{r \in \mathcal{R}} c_r$. Therefore, choosing $\lambda > \max_{r \in \mathcal{R}} c_r$ guarantees that the global minimum of $H(\mathbf{x})$ over $\{0,1\}^{|\mathcal{R}|}$ lies within the feasible region.

Correctness Argument. Minimizing $H(\mathbf{x})$ over binary vectors recovers the optimal solution because (i) for any feasible assignment, all penalty terms vanish, reducing $H(\mathbf{x})$ to the original objective function; (ii) for any infeasible assignment, at least one penalty term is strictly positive. By the choice of λ, the energy increase from constraint violation strictly exceeds any possible decrease in the objective cost, ensuring that the global minimum corresponds to a feasible solution with the minimum delivery cost.

Illustrative Example. Consider a concrete instance with $|\mathcal{J}| = 3$ jobs, $|\mathcal{V}| = 2$ vehicles, and $|\mathcal{R}| = 6$ candidate routes. Let the routes be indexed $r_0, \ldots, r_5$. Jobs $1, 2, 3$ are covered by routes $\{r_0, r_3\}$, $\{r_1, r_4\}$, and $\{r_2, r_5\}$, respectively. Vehicles A and B are assigned routes $\{r_0, r_1, r_2\}$ and $\{r_3, r_4, r_5\}$, respectively. The costs are $c_0 = 5, c_1 = 8, c_2 = 6, c_3 = 7, c_4 = 9, c_5 = 4$. We set $\lambda = 10$, which satisfies $\lambda > \max_r c_r$.

Instantiating the general formulas yields the concrete Hamiltonian:

$$
\begin{aligned}
H(\mathbf{x}) = {} & 5x_0 + 8x_1 + 6x_2 + 7x_3 + 9x_4 + 4x_5 && (11)\\
& + 10(x_0 + x_3 - 1)^2 + 10(x_1 + x_4 - 1)^2 + 10(x_2 + x_5 - 1)^2 && (12)\\
& + 10(x_0x_1 + x_0x_2 + x_1x_2) + 10(x_3x_4 + x_3x_5 + x_4x_5). && (13)
\end{aligned}
$$

Expanding and collecting terms according to the derived coefficients gives:

$$
\begin{aligned}
H(\mathbf{x}) = {} & -5x_0 - 2x_1 - 4x_2 - 3x_3 - 1x_4 - 6x_5 && (14)\\
& + 20x_0x_3 + 20x_1x_4 + 20x_2x_5 + 10x_0x_1 + 10x_0x_2 + 10x_1x_2 && (15)\\
& + 10x_3x_4 + 10x_3x_5 + 10x_4x_5 + 30. && (16)
\end{aligned}
$$

The optimal selection is $x_0 = 1, x_4 = 1, x_5 = 1$ (routes r_0, r_4, r_5), which covers all jobs exactly once and respects vehicle limits. The minimum objective cost is $5 + 9 + 4 = 18$, and the corresponding QUBO energy is $H(\mathbf{x}^*) = 18 + 30 = 48$.